

In the Name of God
The summary of
"7 Revealing Ways AIs Fail"
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This article discusses the critical limitations and vulnerabilities of artificial intelligence (AI), highlighting seven types of failures that impact AI's reliability, safety, and trustworthiness. A key issue is **brittleness**, where AI systems fail when encountering unfamiliar or adversarial examples, exposing their inability to generalize effectively. Another major challenge is **embedded bias**, which stems from biased training data, resulting in discriminatory outcomes in critical decisions, such as healthcare and justice. **Catastrophic forgetting** limits AI's ability to retain knowledge, as systems often overwrite prior learning when updated with new information. The **explainability** problem, where AI decision-making processes remain opaque, undermines trust, especially in high-stakes applications like medical diagnosis. **Quantifying uncertainty** is another weakness; AIs often make overconfident errors without accurately assessing risks, leading to potential disasters in applications like autonomous driving. Furthermore, AIs lack **common sense**, struggling with contextual reasoning and logical inference, which humans typically perform effortlessly. Lastly, AIs demonstrate **mathematical limitations**, performing poorly on tasks requiring sequential or step-by-step reasoning despite their computational power. While researchers are exploring solutions like adversarial training, bias mitigation frameworks, memory retention strategies, and enhanced explainability techniques, these challenges remain significant. Understanding and addressing these failures is essential as AI becomes increasingly integrated into society. Improvements in these areas will help develop more robust, fair, and transparent AI systems that can better serve humanity without unintended consequences.

Ref. Charles Q. Choi. (2021, Oct). 7 Revealing Ways AIs Fail. IEEE Spectrum. [Online]. Available: <https://spectrum.ieee.org/artificial-intelligence-failures>